Examiner only

7. In a fast breeder nuclear reactor, uranium-238 ($^{238}_{92}U$) is converted into the fissionable isotope plutonium-239 ($^{239}_{94}Pu$). The reactor has control rods but no moderator.

The sequence, of the 3 stages, is shown.

1st stage $^{238}_{92}U + ^{1}_{0}n \longrightarrow ^{239}_{92}U$

2nd stage $^{239}_{92}U \longrightarrow ^{239}_{93}Np + ^{0}_{-1}e$

3rd stage $^{239}_{93}Np \longrightarrow ^{239}_{94}Pu + ^{0}_{-1}e$

(a) State the total number of beta particles emitted during the 3 stages. .... [1]

(b) When a plutonium-239 nucleus is bombarded with a **high-speed neutron** it splits into barium (Ba), strontium (Sr) and two high-speed neutrons.
The reaction is shown below.

$$^{239}_{94}Pu + ^{1}_{0}n \longrightarrow ^{145}_{56}Ba + ^{93}_{38}Sr + 2\,^{1}_{0}n$$

The two high-speed neutrons split more plutonium-239 nuclei and cause a nuclear fission chain reaction.
The fast breeder nuclear reactor does not require a moderator, but a conventional nuclear reactor does.

(i) Explain the purpose of the moderator that is used in a conventional nuclear reactor. [2]

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(ii) Suggest why the fast breeder nuclear reactor doesn't require a moderator. [1]

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Examiner only

(c) High level radioactive waste (HLW) from nuclear power stations and nuclear medicine may be contained in solid glass (vitrified) and then kept securely in stainless steel lined, concrete containers in deep underground facilities. HLW is highly ionising and has a long half-life.
Discuss the advantages **and** disadvantages of storing HLW in this way. [6 QER]

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